LAB ASSIGNMENT-6

Aim- Binary Classification using Logistic Regression: Performance Evaluation through Confusion Matrix, Precision, Recall and F1-Score

1. Problem Statement

Illustrate the performance of a binary classification model built using logistic regression on a suitable open-access dataset, and evaluate it through a confusion matrix, precision, recall and F1-score.

Specifically: train a logistic regression classifier to distinguish malignant from benign breast tumours using the Breast Cancer Wisconsin (Diagnostic) dataset, construct the confusion matrix on held-out test data, compute accuracy, precision, recall, specificity and F1-score from its four cells, validate the results with stratified 5-fold cross-validation, and study how the decision threshold shifts the balance between precision and recall.

2. Objectives

- To construct and correctly interpret a confusion matrix in terms of True Positives, True Negatives, False Positives and False Negatives.
- To compute precision, recall, specificity and F1-score from the confusion matrix and verify them against library implementations.

3. Theory

3.1 Logistic Regression

Logistic Regression is a supervised machine learning algorithm mainly used for classification problems, especially when the output has two classes such as Yes/No, Pass/Fail, Spam/Not Spam, or Disease/No Disease. Although its name contains “regression,” it is primarily used for predicting the probability of a class. It uses the sigmoid (logistic) function, which converts the model's output into a value between 0 and 1. For example, if the model gives a probability of 0.85 for “Pass,” it means there is an estimated 85% probability that the student will pass.

The basic idea is to calculate a probability and then use a threshold, commonly 0.5, to make the final classification. If the probability is ≥ 0.5, the sample is classified into class 1; otherwise, it is classified into class 0. Logistic Regression is simple, interpretable, and commonly used for binary classification, although it can also be extended to multiclass classification using approaches such as One-vs-Rest or multinomial logistic regression.

3.2 The Confusion Matrix

A single accuracy number hides the structure of a classifier's errors. The confusion matrix is a 2×2 table cross-tabulating actual against predicted labels, and every standard metric is derived from its four cells

	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0)	True Negative (TN) — correctly identified benign	False Positive (FP) — Type I error: benign flagged as malignant

	Predicted Negative (0)	Predicted Positive (1)
Actual Positive (1)	False Negative (FN) — Type II error: malignant missed	True Positive (TP) — correctly identified malignant

The two off-diagonal cells represent fundamentally different failures with different real-world costs. In this medical application a False Negative is far more serious than a False Positive. No single scalar metric captures this; the matrix must be inspected.

3.3 Performance Metrics

Metric	Formula	Meaning and when to use it
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Overall fraction of correct predictions. Misleading on imbalanced data, and blind to which kind of error occurred.
Precision	TP / (TP + FP)	Of all cases predicted positive, how many really are. Answers: can the alarm be trusted? Maximise when false positives are costly.
Recall (Sensitivity, TPR)	TP / (TP + FN)	Of all actual positives, how many were found. Answers: how many did we miss? Maximise when false negatives are costly, as in disease screening.
Specificity (TNR)	TN / (TN + FP)	Of all actual negatives, how many were correctly cleared. The counterpart of recall for the negative class.
F1-Score	$2 \cdot (P \cdot R) / (P + R)$	Harmonic mean of precision and recall. It is high only when both are high, so it summarises the two in one number for model comparison.

11. Conclusion

It is therefore concluded that logistic regression provides an accurate, interpretable and computationally inexpensive baseline for binary classification, and that its evaluation must be based on the confusion matrix and the metrics derived from it, with the operating threshold selected from the relative cost of the two error types in the intended application.

12. Viva Questions

1. Why logistic regression is called a regression method when it is used for classification?
2. Why can linear regression not be used directly for binary classification?
3. State the sigmoid function and explain the role of the log-odds (logit) transformation.
4. Why is cross-entropy used as the cost function instead of mean squared error?
5. Define True Positive, True Negative, False Positive and False Negative for this experiment.
6. Distinguish precision from recall. Give one application where each should be prioritised.
7. Why is F1-score defined as a harmonic mean rather than an arithmetic mean?